Mohammad Sheriff Mehmood

Senior Data Scientist · NLP & Generative AI Engineer

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Senior Data Scientist & AI/ML Engineer with 6+ Years of experinece.I build AI systems that move business metrics not just demos.Specializing in NLP, LLMs, and production MLOps, I have shipped enterprise-grade GenAI systems that delivered $1M+ revenue uplift, cut churn by 40%, and accelerated model iteration by 60% across healthcare, finance, and retail. Open-source author: published authentica on PyPI, actively used in production at Businessolver (Healthcare Benefits SaaS)

TECHNICAL SKILLS

Languages	Python (primary) proficient in TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, YOLO, CNN
Generative AI	LangChain · LangSmith · OpenAI API · Gemini · RAG · LLMOps · Fine-tuning · Prompt Engineering, Agentic AI,MCP Transformer Models ,Sentence-Transformers,crewAi
MLOps & Cloud	MLflow · Docker · Kubernetes · Airflow · GCP (Vertex AI) · AWS (SageMaker, ECS, CodePipeline) ,CI/CD
Databases	MongoDB ,SQL, PostgreSQL,ChromaDB ,FAISS ,Pinecone
Tools & Other	Tableau · Weights & Biases · FastAPI · Flask · Django · A/B Testing · REST API Design · Git

PROFESSIONAL EXPERIENCE

ValueLabs · Hydrabad, India — Mar 2025 – Present

Senior Data Scientist AI|ML Engineer

Embedded with Businessolver's Data Science team as a full-stack ML Engineer designed and shipped an automated document analysis pipeline leveraging computer vision, VLMs, and AWS-native tooling to process healthcare documents at scale.

- **Document Analysis Pipeline:** Designed an end-to-end multi-modal pipeline on AWS (Comprehend, Textract, S3, ECS) to classify extract data from 47,000+ healthcare documents, significantly cutting manual claim validation time.
- **Query-Based Document Understanding:** Built a custom QA system integrating visual features from Textract OCR with textual embeddings across varied layouts; applied tailored post-processing to produce unified QA outputs and deployed inference on AWS Lambda for scalable processing achieving 93% extraction precision.
- **Annotation & Monitoring UI:** Built an interactive React UI with REST API for data annotation/validation, real-time model metric visualisation, performance monitoring, and data-driven threshold tuning streamlining stakeholder review cycles.
- **Stacking Ensemble Model:** Trained and tuned a stacking ensemble (Random Forest, XGBoost, LightGBM) via cross-validation achieving 0.78 F1 and 0.81 AUC; applied SHAP and Gini impurity for model explainability.Created with react

LTIMindtree · Pune, India — Jan 2024 – Mar 2025

Senior Data Scientist – NLP Engineer

- **GenAI Chatbot & RAG System**: Architected production-grade RAG chatbot (Gemini + FAISS Vector DB) serving 15+ team members with 90% query accuracy and sub-2s response time for enterprise knowledge retrieval.
- **LLMOps Pipeline:** Built end-to-end LLMOps pipeline using OpenAI, LangChain, ChromaDB, and sentence-transformers cutting model iteration time by 60% and reducing deployment overhead by 50%.
- **Healthcare Revenue Model:** Deployed anomaly-detection targeting model generating $1M+ in annual healthcare revenue at 90% precision; reduced false positives by 20% via class-weighted ensemble (Logistic Regression + Random Forest).
- **Workflow Orchestration:** Designed Airflow DAGs to automate multi-step ML pipelines, cutting end-to-end processing time by 40% and improving delivery velocity across a 7-member team.
- **Semantic NLP (BioBERT):** Generated vector embeddings for ICD-9 and medical procedure codes using BioBERT, improving semantic search accuracy by 18% over TF-IDF baseline.

Cognizant Technology Solutions - Chennai, India — Dec 2019 – Jan 2024

Data Scientist

Joined as Programmer Analyst core ML systems, promoted to Data Scientist (leading cross-functional AI/NLP initiatives.

- **ML Model Development:** Built and deployed supervised ML models improving prediction accuracy by 20% and saving. saving $200K in operational costs; improved baselines by 5–10% using Neural Networks, ensemble methods and GridSerachCV,hyperparameter tunning
- **AI Customer Support Platform:** Led 7-member cross-functional team to build NLP-powered support application; reduced customer churn by 40% and generated $300K additional quarterly revenue through context-aware intelligent responses.
- **Fraud Detection System:** Reduced false-positive fraud alerts by 20% using Logistic Regression and Random Forest with class-weighted resampling — received company Innovation Award.
- **Data Platform & BI:** Unified 10+ heterogeneous data sources (databases, APIs, third-party feeds) into ML-ready pipelines; built Tableau dashboards tracking 20+ KPIs, improving cross-team reporting efficiency by 30%.

RESEARCH PROJECTS

Accident & Fall Detection System — View on GitHub →

Python · YOLO · OpenCV · CNN · CUDA · SMTP

- Built real-time human fall detection system achieving 95% accuracy using YOLO + CUDA toolkit on edge hardware.
- Designed smart camera surveillance for highway accident detection; automated ambulance dispatch via SMTP alerts, cutting emergency response time by 40%.

Abstractive Text Summarisation — View on GitHub →

Python · BERT · NLP · Transformers

- Applied BERT and NLP toolkits to summarise live chat context, improving agent efficiency by 35% and reducing customer response times by 25%.
- Gave agents instant access to previous chat history — reducing handle time and improving resolution quality.

Visa Approval Forecasting (US Immigration) — View on GitHub →

Python · ML · MongoDB · Docker · AWS · GridSearchCV

- Predicted US visa approval outcomes using production-grade ML models; performed EDA, feature engineering, and hyperparameter tuning via GridSearchCV.
- Connected MongoDB for data persistence; containerised solution with Docker for AWS deployment.

Disease Classification Mobile App — View on GitHub →

TensorFlow · CNN · TFLite · FastAPI · GCP · ReactJS · ReactNative

- Achieved 92% accuracy detecting Early/Late Blight in potato crops; quantized CNN with TF Lite — reduced model size by 80% and inference latency by 50% for on-device deployment.
- Deployed serverless inference on Google Cloud Functions (25% cost reduction); served real-time predictions via TF Serving + FastAPI to ReactJS and React Native frontends.

End-to-End Topic Modelling & MLOps on AWS — View on GitHub →

Python · LDA · Docker · AWSECS · MLflow · CodePipeline · EC2

- Built LDA topic modelling pipeline on AWS ECS with blue/green deployment — cut processing time by 40% and deployment time by 60%.
- Automated full CI/CD (CodeCommit, CodeBuild, CodeDeploy, CodePipeline); reduced manual effort by 70%. Used MLflow for model versioning; ensured high availability via EC2 + ECS load balancing.

EDUCATION

M.Tech in Data Science & Engineering · BITS Pilani, India — Oct 2021 – Oct 2023

B.E in Computer Science & Engineering ·OIST Bhopal, India — Aug 2015 – Jun 2019

CERTIFICATIONS & AWARDS

1st Prize – Trizetto Hackathon (Patient Data Integration, 95% approval rate)

GCP Professional ML Engineer (Certified Feb 2023)

Certified MLOps Developer – Dataiku

IBM Professional Data Science Certification

Secretary, Toastmasters Club · Speaker, PyData Conference

OpenAI Whisper Hackathon – Hearing-impaired accessibility app